

TRIGGERS

Scenario 1: UpdateCustomerLastModified

```
CREATE OR REPLACE TRIGGER UpdateCustomerLastModified
BEFORE UPDATE
ON Customers
FOR EACH ROW
BEGIN
    :NEW.LastModified := SYSDATE;
END;
/
```

The screenshot displays the FreeSQL web application interface. At the top, there are browser tabs for 'Cognizant GenC Ne...', 'Free JavaScript Cour...', 'Uninstall VPN Brow...', and 'Cognizant Skillspring'. The FreeSQL interface includes a 'Navigator' panel on the left with a tree view showing 'My Schema' and 'Tables'. Below this is a search bar and a list of database objects: ACCOUNTS, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main workspace is titled '[SQL Worksheet]*' and contains the SQL code for creating the trigger. Below the code editor, there are tabs for 'Query result', 'Script output', 'DBMS output', 'Explain Plan', and 'SQL history'. The 'Script output' tab is active, showing the execution of the SQL statement and the message 'Trigger UPDATECUSTOMERLASTMODIFIED compiled'. The elapsed time is 00:00:00.018.

```
1 CREATE OR REPLACE TRIGGER UpdateCustomerLastModified
2 BEFORE UPDATE
3 ON Customers
4 FOR EACH ROW
5 BEGIN
6     :NEW.LastModified := SYSDATE;
7 END;
8 /
```

SQL> CREATE OR REPLACE TRIGGER UpdateCustomerLastModified
BEFORE UPDATE
ON Customers
FOR EACH ROW...
[Show more...](#)

Trigger UPDATECUSTOMERLASTMODIFIED compiled

Elapsed: 00:00:00.018

UPDATE Customers
SET Balance = Balance + 1000
WHERE CustomerID = 1;

COMMIT;

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FreeSQL >_ Worksheet SQL Tutorials

Navigator Files

My Schema

Tables

Search

ACCOUNTS

CUSTOMERS

EMPLOYEES

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TRANSACTIONS

[SQL Worksheet]*

```
1 UPDATE Customers
2 SET Balance = Balance + 1000
3 WHERE CustomerID = 1;
4
5 COMMIT;
6
```

Query result Script output DBMS output Explain Plan SQL history

1 row updated.

Elapsed: 00:00:00.012

SQL> COMMIT

Commit complete.

Elapsed: 00:00:00.003

Query result Script output DBMS output Explain Plan SQL history

Download Execution time: 0.137 seconds

	CUSTOMERID	NAME	BALANCE	LASTMODIFIED
1	1	John Doe	2000	7/17/2026, 3:16:38

Scenario 2: LogTransaction

```
CREATE TABLE AuditLog (  
    LogID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
    TransactionID NUMBER,  
    AccountID NUMBER,  
    Amount NUMBER,  
    TransactionType VARCHAR2(20),  
    LogDate DATE  
);
```

The screenshot displays the FreeSQL SQL Worksheet interface. On the left, the 'Navigator' pane shows a schema named 'My Schema' containing several tables: ACCOUNTS, AUDITLOG, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main workspace, titled '[SQL Worksheet]*', contains the following SQL code:

```
1 CREATE TABLE AuditLog (  
2     LogID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY  
3     TransactionID NUMBER,  
4     AccountID NUMBER,  
5     Amount NUMBER,  
6     TransactionType VARCHAR2(20),  
7     LogDate DATE  
8 );
```

Below the code editor, the 'Script output' tab is active, showing the execution results:

```
SQL> CREATE TABLE AuditLog (  
      LogID NUMBER GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,  
      TransactionID NUMBER,  
      AccountID NUMBER,...  
Show more...
```

Table AUDITLOG created.

Elapsed: 00:00:00.022

```

CREATE OR REPLACE TRIGGER LogTransaction
AFTER INSERT
ON Transactions
FOR EACH ROW
BEGIN
    INSERT INTO AuditLog(
        TransactionID,
        AccountID,
        Amount,
        TransactionType,
        LogDate
    )
    VALUES(
        :NEW.TransactionID,
        :NEW.AccountID,
        :NEW.Amount,
        :NEW.TransactionType,
        SYSDATE
    );
END;
/

```

The screenshot displays the FreeSQL web-based SQL editor. The interface includes a sidebar with a 'Navigator' pane showing a database schema with tables: ACCOUNTS, AUDITLOG, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main editor area, titled '[SQL Worksheet]*', contains the SQL script for creating the LogTransaction trigger. Below the editor, the 'Script output' pane shows the successful execution of the script, including a confirmation message and the elapsed time.

```

1  CREATE OR REPLACE TRIGGER LogTransaction
2  AFTER INSERT
3  ON Transactions
4  FOR EACH ROW
5  BEGIN
6      INSERT INTO AuditLog(
7          TransactionID,
8          AccountID,
9          Amount,
10         TransactionType,
11         LogDate
12     )
13     VALUES (
14         :NEW.TransactionID,
15         :NEW.AccountID,
16         :NEW.Amount,
17         :NEW.TransactionType,

```

Query result **Script output** DBMS output Explain PI

SQL> CREATE OR REPLACE TRIGGER LogTransaction
AFTER INSERT
ON Transactions
FOR EACH ROW...

Trigger LOGTRANSACTION compiled

Elapsed: 00:00:00.020

Query result
Script output
DBMS output
Explain Plan
SQL his

1 row inserted.
Elapsed: 00:00:00.024

SQL> COMMIT

Commit complete.

Query result

Script output

DBMS output

Explain Plan

SQL history

Download

Execution time: 0.145 seconds

	LOGID	TRANSACTIONID	ACCOUNTID	AMOUNT	TRANSACTIONTYPE	LOGDATE
1	1	301	101	500	Deposit	7/17/2026, 3:20:40

Scenario 3: CheckTransactionRules

```

CREATE OR REPLACE TRIGGER CheckTransactionRules
BEFORE INSERT
ON Transactions
FOR EACH ROW
DECLARE
    v_Balance NUMBER;
BEGIN
    SELECT Balance
    INTO v_Balance
    FROM Accounts
    WHERE AccountID = :NEW.AccountID;

    IF :NEW.TransactionType = 'Withdrawal' THEN
        IF :NEW.Amount > v_Balance THEN

```

```

        RAISE_APPLICATION_ERROR(
            -20010,
            'Withdrawal exceeds available balance.'
        );
    END IF;
ELSIF :NEW.TransactionType = 'Deposit' THEN
    IF :NEW.Amount <= 0 THEN
        RAISE_APPLICATION_ERROR(
            -20011,
            'Deposit amount must be positive.'
        );
    END IF;
END IF;
END;
/

```

Query result	Script output	DBMS output	Explain Plan	SQL history
  <pre> SQL> CREATE OR REPLACE TRIGGER CheckTransactionRules BEFORE INSERT ON Transactions FOR EACH ROW... Show more... Trigger CHECKTRANSACTIONRULES compiled Elapsed: 00:00:00.024 </pre>				

Valid Deposit:

```

INSERT INTO Transactions(
    TransactionID,
    AccountID,
    TransactionDate,
    Amount,
    TransactionType
)

```

```
VALUES(  
    302,  
    101,  
    SYSDATE,  
    1000,  
    'Deposit'  
);
```

```
COMMIT;
```

Query result	Script output	DBMS output	Explain Plan	SQL history
 				
1 row inserted.				
Elapsed: 00:00:00.016				
SQL> COMMIT				
Commit complete.				
Elapsed: 00:00:00.002				

Invalid Withdrawal:

```
INSERT INTO Transactions(  
    TransactionID,  
    AccountID,  
    TransactionDate,  
    Amount,  
    TransactionType  
)  
VALUES(  
    303,  
    101,  
    SYSDATE,  
    50000,
```

'Withdrawal'
);

The screenshot shows the FreeSQL SQL Worksheet interface. The left sidebar contains a 'Navigator' pane with 'My Schema' and 'Tables' dropdowns, a search bar, and a list of database objects: ACCOUNTS, AUDITLOG, CUSTOMERS, EMPLOYEES, LOANS, and TRANSACTIONS. The main editor area displays an SQL statement:

```
1  INSERT INTO Transactions(  
2      TransactionID,  
3      AccountID,  
4      TransactionDate,  
5      Amount,  
6      TransactionType  
7  )  
8  VALUES (  
9      303,  
10     101,  
11     SYSDATE,  
12     50000,  
13     'Withdrawal'  
14 );
```

Below the editor, the 'Script output' tab is active, showing the execution results:

```
SQL> INSERT INTO Transactions(  
      TransactionID,  
      AccountID,  
      TransactionDate,...  
Show more...
```

The output also displays error messages:

```
ORA-20010: Withdrawal exceeds available balance.  
ORA-06512: at "PRERNAP1067_SSE_SCHEMA_4GL2E.CHECKTRANSACTIONRULES", line 11  
ORA-04088: error during execution of trigger 'PRERNAP1067_SSE_SCHEMA_4GL2E.CHECKTRANSACTIONRULES'  
  
https://docs.oracle.com/error-help/db/ora-20010/  
Error at Line: 4 Column: 0
```

Invalid Deposit:

```
INSERT INTO Transactions(  
    TransactionID,  
    AccountID,  
    TransactionDate,  
    Amount,  
    TransactionType  
)  
VALUES(  
    304,  
    101,  
    SYSDATE,  
    -100,
```

'Deposit'
);

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FreeSQL Worksheet SQL Tutorials

Navigator Files

My Schema

Tables

Search

ACCOUNTS

AUDITLOG

CUSTOMERS

EMPLOYEES

LOANS

TRANSACTIONS

```
1  INSERT INTO Transactions(
2      TransactionID,
3      AccountID,
4      TransactionDate,
5      Amount,
6      TransactionType
7  )
8  VALUES (
9      304,
10     101,
11     SYSDATE,
12     -100,
13     'Deposit'
14 );
```

Query result Script output DBMS output Explain Plan SQL history

SQL> INSERT INTO Transactions(
TransactionID,
AccountID,
TransactionDate,...
Show more...

ORA-20011: Deposit amount must be positive.
ORA-06512: at "PRERNAP1067_SSE_SCHEMA_4GL2E.CHECKTRANSACTIONRULES", line 18
ORA-04088: error during execution of trigger 'PRERNAP1067_SSE_SCHEMA_4GL2E.CHECKTRANSACTIONRULES'
<https://docs.oracle.com/error-help/db/ora-20011/>
Error at Line: 4 Column: 0